

A Comprehensive Approach to Transient Stability Control: Part I—Near Optimal Preventive Control

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Abstract—A general approach to real-time transient stability control is proposed, and two complementary techniques are devised: one for preventive, the other for emergency control. In this paper, the general transient stability control approach is first revisited then applied to real-time preventive control. The technique consists of shifting active power generation. The amount of power and the machines from which to shift it are methodically determined, and various patterns of generation decrease/increase are considered. Further, a standard OPF algorithm is combined with this control technique to get a real-time transient stability-constrained OPF software, able to meet power system security and electricity market requirements. Simulations conducted on the 88-machine EPRI system and the Mexican interconnected power system illustrate the various techniques, highlight their specifics, and assess their performance.

Index Terms—DSA, preventive control, SIME method, transient stability, transient stability control, transient stability-constrained OPF.

I. INTRODUCTION

POWER system dynamics has long been recognized as an important and problematic issue. The electric market liberalization increases its importance, as economical pressure and intensified transactions tend to operate electric power systems much closer to their security limits than ever before. At the same time, it aggravates the difficulties, given the trend to merge existing systems into much larger entities and to monitor them in shorter and shorter time horizons. This holds true for analysis aspects and even more for control, inasmuch as, today, control actions must cope with considerably more stringent market requirements than in the past. The difficulties increase further when it comes to transient stability phenomena, which develop very fast and whose control involves active power.

This two-part paper proposes a general transient stability control method, from which it derives techniques able to provide near-optimal control countermeasures, while encountering electricity market requirements, in two distinct ways. The one, devoted to real-time preventive control, is elaborated in this first part paper; the other, devoted to open-loop emergency control, is addressed in the companion paper [1].

The various techniques rely on the single machine equivalent (SIME) hybrid transient stability method. A detailed description of SIME may be found in [2], while a short account is given in [3]. Basically, SIME drives a time-domain (T-D)

transient stability program in order to replace the dynamics of the multimachine power system by that of a suitable one-machine infinite bus (OMIB) system. By refreshing the OMIB parameters at each step of the T-D program and by assessing the OMIB stability by the equal-area criterion, SIME provides accurate and fast transient stability assessment. Even more important is that SIME calculates stability margins and identifies critical machines, which are the core of the proposed generation rescheduling technique.

Indeed, generation rescheduling aims at shifting generation from selected machines. Its amount depends on the size of the stability margin, while the machines from which it should be shifted are critical machines. Note that the generation rescheduling may be realized in various ways, depending upon the objective(s) sought. Note also that by combining a conventional OPF algorithm with the proposed control technique allows meeting steady-state and transient stability constraints together with additional objectives dictated by the electricity market. Besides, the resulting solutions may be transparent to the market participants. In all cases, the basic preventive control technique, combined or not with OPF, provides a wide range of near-optimal solutions to various requirements.

This paper revisits the fundamentals of the SIME-based control to derive preventive control techniques. It then scrutinizes features and performance of these techniques by simulations conducted on the EPRI 88-machine system [4] and the Mexican interconnected system, modeled in their usual, detailed way [5]. In all of these simulations, the techniques use the SIME software coupled with the ETMSP time-domain program [6].

II. FUNDAMENTALS OF TRANSIENT STABILITY CONTROL

A. SIME: Fundamentals and Direct Products

NB. This section revisits material taken from [2], [3], in order to introduce notation and to make the paper self-reliant. The illustrations are obtained from simulations performed in subsequent sections of this and the companion papers.

To analyze an unstable case (defined by the prefault system operating conditions and the contingency scenario), SIME starts driving a time-domain (T-D) program as soon as the system enters its postfault configuration. At each step of the T-D simulation, SIME transforms the multimachine system furnished by this program into a suitable one-machine infinite bus (OMIB) equivalent, defined by its angle δ , speed ω , mechanical power P_m , electrical power P_e , and inertia coefficient M . (All OMIB parameters are derived from multimachine system parameters.) Further, SIME explores the OMIB dynamics by using the equal-area criterion (EAC). The procedure stops

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as soon as the OMIB reaches the EAC **instability conditions** expressed by

$$P_a(t_u) = 0; \quad \dot{P}_a(t_u) > 0 \quad (1)$$

where P_a is the OMIB accelerating power, difference between P_m and P_e , and t_u is the **time to instability**: at this time, the OMIB system loses synchronism, and the system machines split irrevocably into two groups: the group of “advanced machines” that we will henceforth refer to as the “**critical machines**” (CMs), and the remaining ones, called the “**noncritical machines**,” (NMs)¹. Thus, at t_u SIME determines

- the CMs, responsible of the system loss of synchronism;
- the stability margin:

$$\eta_u = A_{dec} - A_{acc} = -\frac{1}{2}M\omega_u^2. \quad (2)$$

Figs. 1–3 illustrate the above definitions on a stability case of the EPRI system, simulated in Section III. More precisely, Fig. 1 portrays the multi-machine swing curves provided by the ETMSP program, stopped respectively at “the time to instability”, t_u , (unstable case) and at the maximum integration period (5 s; stable case). The symbol P_C used in the captions denotes the total power of the group of critical machines. Time t_r , indicated in Fig. 1(b), is defined by the **stability conditions**

$$\omega(t_r) = 0; \quad P_a(t_r) < 0. \quad (3)$$

At t_r , the first-swing stability margin can be computed by

$$\eta_{st} = \int_{\delta_r}^{\delta_u} P_a d\delta. \quad (4)$$

Fig. 2 portrays the equivalent OMIB swing curves, while Fig. 3 represents, in the δ - P plane, the evolution with δ of powers P_m and P_e . To simplify, Fig. 3(b) displays only part of the evolution of the electrical power P_e .

B. SIMEs Salient Parameters and Properties

- 1) Calculation of stability margins, identification of the critical machines, and assessment of their degree of criticality (or participation to the instability phenomena) are parameters of paramount importance.
- 2) The “**time to instability**” t_u is another important factor. It indicates the time an unstable simulation is stopped, and measures its severity.
- 3) Similarly, the “**time to first-swing stability**” indicates the time where the system is identified as first-swing stable. If multiswing instabilities are not of concern, it can be used as an early termination criterion for stable simulations.
- 4) The margin expressed by (2) is often “normalized” by the OMIB inertia coefficient. In what follows, we will refer to this latter as the “standard” stability margin.
- 5) Under very unstable conditions, it may happen that the standard margin does not exist, because the OMIB P_m and P_e curves do not intersect (there is no postfault equilibrium solution). A convenient substitute is the “min-

¹The advanced machines are the CMs for upswing instability phenomena, while for backswing phenomena they become NMs.

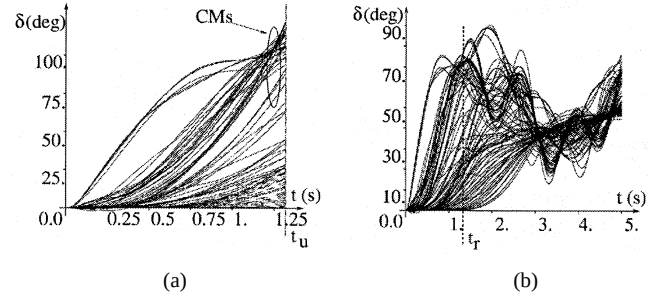


Fig. 1. Multimachine swing curves. Ctg # 11 applied to the EPRI system. Critical clearing time: 69 ms; clearing time: 95 ms. (a) Unstable case: $P_c = 26.162$ MW. (b) Stable case: $P_c = 25.377$ MW.

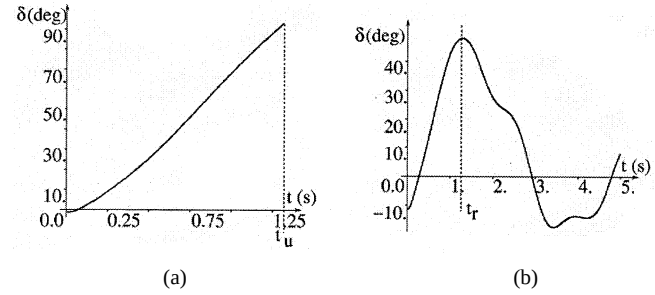


Fig. 2. OMIB swing curves. Ctg # 11 applied to the EPRI system. (a) Unstable case: $t_u = 1.315$ s. (b) Stable case: $t_r = 1.285$ s.

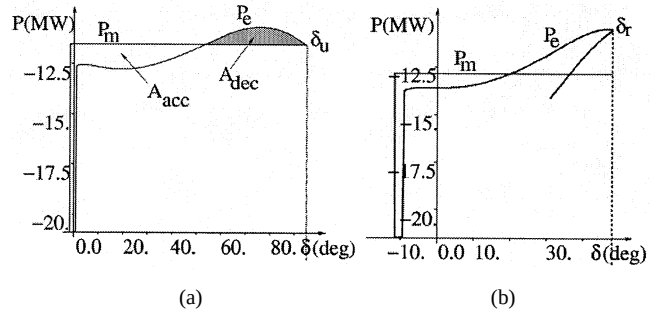


Fig. 3. OMIB δ - P curves. Ctg # 11 applied to the EPRI system. (a) Unstable case: $\delta_u = 92$ degrees. (b) Stable case: $\delta_r = 48.11$ degrees.

imum distance” between postfault P_m and P_e curves. Fig. 4 illustrates this substitute margin under the particularly stressed conditions simulated in Section III. Note that here the “time to instability” is the time to reach this minimum distance and to stop the simulation. To simplify, we will still denote it “ t_u .”

- 6) A very interesting general property of the stability margins (standard as well as substitute ones) is that they vary quasi-linearly with the stability conditions [2]. Fig. 5 illustrates the margin variation with the total generation power of the group of critical machines P_C . The proposed control techniques benefit considerably from this property.
- 7) “**Seconds of T-D integration**” (sTDI). Given that the computing effort required by SIME per se is virtually negligible as compared with the computation of T-D simulations, sTDI is a handy “measure” of computing effort: it represents the time required by the T-D program to run the simulation, and thus, renders comparisons of computing

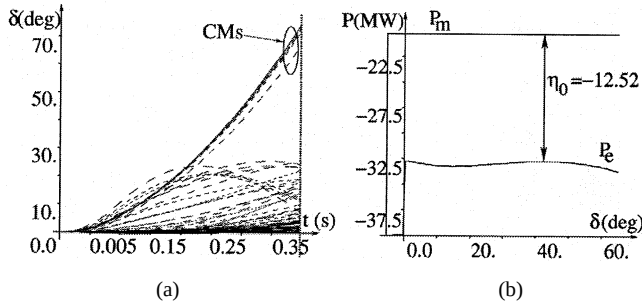


Fig. 4. Example of a severely stressed case: Ctg # 1b applied to the EPRI system. Clearing time = 95 ms; $P_C = 5600$ MW. (a) Multimachine swing curves. (b) OMIB δ - P curves.

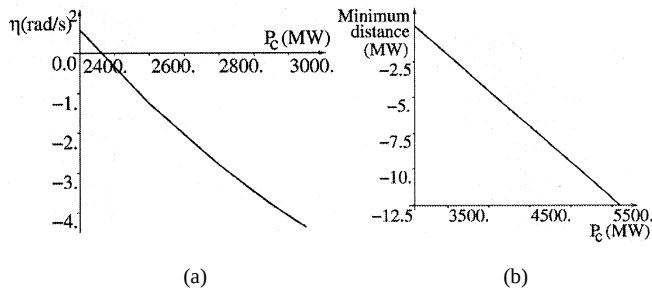


Fig. 5. Typical variation of the standard stability margin and of its substitute with P_C . Drawn on the EPRI system. Ctg. # 1b. (a) Stability margin versus P_C . (b) Minimum distance versus P_C .

performance (almost) independent of the computer used and of the system size.

C. Principle of SIME-Based Transient Stability Control

To stabilize an unstable case (defined by the prefault operating conditions and the contingency type and clearing scenario), SIME furnishes the following two-part information.

- **Size of instability (margin) and critical machines** along with their **degree of criticality or involvement**; the degree of involvement of a critical machine is proportional to its angular deviation d assessed at t_u .
- **Suggestions for stabilization.** These suggestions are obtained by the interplay between OMIB-EAC (equal-area criterion) and time-domain multimachine representations, according to the following procedure:
 - Stabilizing an unstable case consists of modifying the precontingency conditions until the stability margin becomes zero. According to EAC, this implies increasing the decelerating area and/or decreasing the accelerating area of the OMIB δ - P representation. Fig. 3(a) suggests that, in turn, this may be achieved by decreasing the OMIB equivalent generation power. Reference [2] derives a relationship between the margin η and the amount of the OMIB generation decrease ΔP_{OMIB}

$$\eta = f(\Delta P_{OMIB}). \quad (5)$$

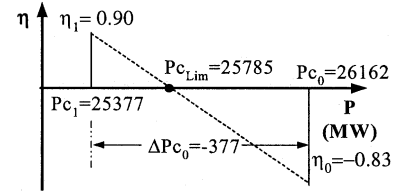


Fig. 6. SIME-based stabilization procedure. Ctg #11; conditions of Fig. 1.

- Further, [2] shows that to keep the total consumption constant, the following multimachine condition must be satisfied, when neglecting losses:

$$\begin{aligned} \Delta P_{OMIB} &= \Delta P_C = \sum_{i \in CMs} \Delta P_{Ci} = -\Delta P_N \\ &= - \sum_{j \in NM_s} \Delta P_{Nj} \end{aligned} \quad (6)$$

where ΔP_C and ΔP_N are the changes in the total power of the group of critical and noncritical machines, respectively.

- Application of (5) and (6) provides a first approximate value of ΔP_C .

Remark: Obviously, generation redispatch is corroborated by engineering reasoning: to stabilize a system, bring the machines' angle trajectories closer to each other. However, SIME provides important additional information: it quantifies the amount of generation to be shifted and determines the machines from which it should be shifted.

III. PREVENTIVE CONTROL

A. Iterative Stabilization Procedure

The reasoning of Section II-C yields the iterative stabilization procedure described in Fig. 6 and summarized below.

For a given negative (unstable) margin η_0

- decrease the total initial power of the critical machines P_{C0} by ΔP_{C0} to get the new power $P_{C1} = P_{C0} - \Delta P_{C0}$, using either (5) and (6) or a small percentage of the initial generation;
- increase by the same amount the total active power of the noncritical machines;
- perform successively a power flow to compute the new operating conditions, and a stability run to compute the corresponding stability margin η_1 , and
 - if $\eta_1 = 0$, stop
 - otherwise, perform a linear interpolation (or extrapolation as appropriate), to get a first-guess power limit P_{CLim} , then go to step (i) and decrease or increase the active power of the critical machines, as appropriate.

B. Generation Rescheduling Patterns

Expression (6) suggests that there exist numerous ways of distributing the total power change ΔP_N among noncritical machines. In addition, whenever there are many critical machines,

TABLE I
STABILIZATION OF CTG # 1b [5]

1	2	3	4	5	6
It. #	η (rad/s) ²	Nr of CMs	P_{Ck} (MW)	P_{Ck+1} (MW)	sTDI (s)
0	-12.52*	7	5600	5041	0.395
1	-9.70*	7	5041	3431	0.430
2	-1.66*	7	3431	2815	0.605
3	-2.898	7	2815	2731	0.935
4	-2.282	7	2731	2423	1.015
5	0.675	7	2423	2493	5.000

there exist numerous patterns for distributing ΔP_C among them [2], [3], [7].

Concerning distribution of generation among noncritical machines, the patterns may be dictated by various objectives, related to market or technical considerations; for example, one may seek for maximum transfer capability on a given corridor.

In the absence of particular constraints or objectives, the total generation power could be distributed proportionally to the inertias of the noncritical machines. Another, interesting solution consists of using an optimal power flow (OPF) program (see Section IV).

Further, instead of stabilizing individually each “unstable contingency” one can design a pattern able to stabilize all harmful contingencies simultaneously (see Section III-D).

Below, we illustrate the two alternatives by simulations performed on the EPRI-88 machine system (434 buses, 2357 lines) [4]. The base case (prefault operating condition) has a total generation of 350 749 MW. It is subjected to each one of the 36 contingencies proposed in [4], which correspond to three-phase short-circuits applied at 500-kV buses and cleared 95 ms after their occurrence, generally by opening one or several lines.

Out of the 36 contingencies, the FILTRA technique, described in [3], has identified five harmful (or “unstable”) contingencies, labeled # 1a, 1b, 10, 11, and 30.²

C. Single Contingency Stabilization

Fig. 6 described the simulations performed to stabilize ctg # 11, which creates a moderate instability.

Table I describes the procedure for stabilizing ctg # 1b, which creates a very severe instability. Asterisk in column 2 indicates a substitute margin (see item 5 in Section II-B and Fig. 4(b)). In such a case, the iterative procedure starts with substitute margins until curves P_e and P_m intersect, then continues with standard margins. Note that, generally, such severe cases require a larger number of iterations.

The individual stabilization of the above mentioned five harmful contingencies is summarized in Table II, where the generation shift is equally distributed on all critical machines (CMs) and redispatched on noncritical machines (NMs) proportionally to their inertias. Note that both indicators, initial margin, and time to instability, provide the same contingency severity ranking: ctg # 1b, 30, 1a, 11, 10. Incidentally, this

²When clearing the fault, Ctg# 1a, 10 and 11 trip one line while Ctg# 30 and 1b trip two lines. A more detailed description of the contingencies can be found in [4]. Note that power system operating conditions where five credible contingencies result in multiple plant loss-of-synchronism represents very high risk operating conditions that are not considered to be an acceptable practice in the industry. As shown by the application examples, the present method is able to help the operator identify and solve this problem online.

TABLE II
INDIVIDUAL STABILIZATION OF HARMFUL CONTINGENCIES

Cont. Nr	P_{C0} (MW)	ΔP_C (MW)	Nr of CMs	η_0 (rad/s) ²	t_{w0} (ms)	Nr of It.	sTDI
1b	5600	3178	7	-12.52*	0.33	6	8.375
30	27014	804	38	-1.21	0.91	2	5.91
1a	24623	627	32	-0.88	1.19	2	6.19
11	26162	325	36	-0.83	1.31	2	6.31
10	26162	325	36	-0.69	1.34	2	6.34

TABLE III
SIMULTANEOUS STABILIZATION OF FOUR HARMFUL CONTINGENCIES

1	2	3	4	5
	Cont. # 1a	Cont. # 10	Cont. # 11	Cont. # 30
Iteration 0				
η (rad/s) ²	-0.88	-0.69	-0.83	-1.21
P_{C0}	24623	26162	26162	27014
Nr of CMs	32	36	36	38
Machine Nr	Pcoi	Angle order (di)	Angle order (di)	Angle order (di)
1877 821	1 (66.5)	21 (53.3)	19 (52.4)	2 (79.4)
1878 769	2 (66.4)	23 (52.9)	21 (51.9)	1 (79.2)
1873 821	3 (66.3)	22 (52.0)	20 (50.9)	4 (79.1)
1870 821	4 (66.3)	19 (46.7)	18 (47.9)	3 (79.1)
1771 779	5 (63.1)	24 (44.8)	26 (47.9)	5 (74.8)
1855 821	6 (63.1)	2 (44.7)	24 (46.6)	6 (74.5)
1854 769	7 (61.2)	30 (43.6)	30 (46.0)	7 (74.2)
Iteration 1				
η (rad/s) ²	0.249	1.223	1.218	0.001
P_{C1}	23812	25351	25351	26210
(limit)	(23994)	(25870)	(25832)	(26203)

ranking is also in agreement with CMs’ total generation decrease. Finally, observe that among the five harmful contingencies, ctg#1b is, by far, the severest: its stabilization imposes a generation decrease in the seven CMs of over 50% of their total initial generation (versus 1.44% in the case of ctg # 11).

D. Simultaneous Stabilization of Many Contingencies

1) *Principle*: The principle is simple: to stabilize simultaneously many harmful contingencies having in common (some) CMs, apply the most constraining power shift, imposed by the severest contingency, to these common CMs only. The simultaneous stabilization may comply with various patterns, just like individual stabilization.

As an example, Table III describes the simultaneous stabilization of the four less harmful contingencies (# 30, 1a, 11, and 10), when distributing the generation shift equally on their seven common CMs listed in column 1. Observe that a total generation shift of $27,014 - 26,210 = 804$ MW is sufficient to stabilize the system against ctg # 30 and, hence, against the other three milder contingencies (actually these latter are slightly overstabilized, as suggested by the positive margins of iteration 1).

Finally, Table IV lists the power shift from CMs (ΔP_C), the number of iterations and the corresponding number of sTDI required to stabilize simultaneously the four “mild” contingencies and all five contingencies, respectively.

2) Observations and Comments:

1) In terms of computing times, observe that stabilizing many contingencies simultaneously is at least as fast as stabilizing them individually, in a sequence. [Compare the number of sTDI of last column of Table IV with the corresponding sum of the sTDI of the (four or five) contingencies of Table II]. Observe also that the computing

TABLE IV
SIMULTANEOUS STABILIZATION PROCEDURE

P_{c0} (MW)	$P_{c,final}$ (MW)	ΔP_c (%)	ΔP_c (MW)	Nr of It.	sTDI
Simultaneous stabilization of 4 ctgs (excluding 1b)					
27014	26210	2.97	804	1	24.75
Simultaneous stabilization of all 5 ctgs					
5600	2423	56.73	3177	5	26.82

time would be reduced significantly if the exploration were limited to first-swing stability (the exploration of stable cases would have been stopped at the “return time,” rather than extended to 5 s).

Assuming that 1 sTDI corresponds to a few seconds of CPU time suggests that the computing times of the transient stability control software may easily comply with online requirements be it for individual or simultaneous contingencies stabilization.

- 2) In terms of total generation decrease in CMs, the simultaneous stabilization is quite similar with the individual ones, provided that the severity of the various contingencies is about the same. If so, the generation shift is rather inexpensive (the decrease in generation of the CMs is a small percentage of their total initial generation).

However, this stops being true when stabilizing a very harmful contingency, like ctg # 1b. The last row of Table IV shows that the countermeasure becomes very expensive and, actually, unacceptable by the system operator. One is then led to resort to alternative solutions, like the open-loop emergency control proposed in the companion paper [1].

- 3) Generally, the number of sound patterns is very large; this provides many possible choices for CMs during the stabilization procedure. A significantly larger number of choices exist for NMs. Many solutions may therefore be exploited in order to achieve preassigned objectives. The transient stability-constrained OPF technique described below is shown to provide interesting solutions.

IV. TRANSIENT STABILITY-CONSTRAINED OPF

A. Traditional Scope of OPF

Optimal power flow (OPF) uses control variables like active and reactive generation powers to achieve a good tradeoff between security and economics [8]. More specifically, this program optimizes the power system operating condition with respect to a prespecified objective (minimum operating cost, maximum power flow), while respecting generator limits and static security constraints (line power flow and bus voltage limits).

B. Revisiting Transient Stability-Constrained OPF

Several attempts have been made to imbed transient stability constraints within OPF. According to the way of handling these constraints, they yielded two different approaches that below we call “global” and “sequential.”

- **Global approach.** A time-domain (T-D) simulation is run. The power system transient stability model is converted

into an algebraic set of equations for each time step of this simulation. The set of nonlinear algebraic equations resulting from the whole T-D simulation is then included in the OPF as a stability constraint, forming a (generally huge) single nonlinear programming problem (e.g., see [9] and [10]).

- **Sequential approach.** A T-D simulation is run. The transient stability constraints are directly converted into conventional constraints of a standard OPF program (e.g., active generation power). Hence, they do not affect the size of the power system model and the complexity of the OPF solution method. They can use any conventional OPF program (e.g., see [5] and [11]–[13]).

Conceptually, the global approach is more appealing: it is supposed to handle the problem as a whole, and hence, to provide an optimal solution, which would be accepted as the reference by the system operator and the electric market participants. However, it also raises a few objections: it lacks transparency about the salient parameters responsible for the system loss of synchronism, and the reasons underlying the advocated solution; it does not propose alternative solutions; it requires heavy computations due to the huge programming model; it generally uses simplified power system modeling in order to make the whole procedure compatible with acceptable computing requirements. Further, in very stressed systems where modeling details are necessary for assessing correctly power system limits, convergence problems can also arise because the additional constraints, modeled as a large set of algebraic equations, may be ill conditioned. Finally, increasing the number of constraints treated by the global function might result in overly conservative stability assessment. All in all, many causes of suboptimality are generally hidden behind the theoretically proven optimality.

As concerning the sequential approach, the main objection is that it cannot guarantee optimality.

C. SIME-Based Sequential Approach

In principle, the SIME-based transient stability-constrained techniques may comply with either of the above approaches. We have preferred the sequential approach for above advocated reasons and for the following additional ones. i) While in theory this approach cannot guarantee optimality, in practice it provides near optimal solutions, thanks to SIME’s multifaceted information (detection of the very moment the system loses synchronism; identification of the CMs; computation of stability margins; determination of the amount of power to shift). ii) The sequential approach may easily comply with market requirements thanks to the flexibility of choice among CMs and NMs on which generation can be redispatched. iii) The approach can easily be adapted to auction market software different from the OPF program.

Fig. 7 illustrates the principle of the sequential approach. Its various steps are summarized as follows.

- i) The TSA block identifies the harmful contingencies and sends the appropriate information (operating conditions along with CMs and margins) to the transient stability control (TSC) block for stabilization.

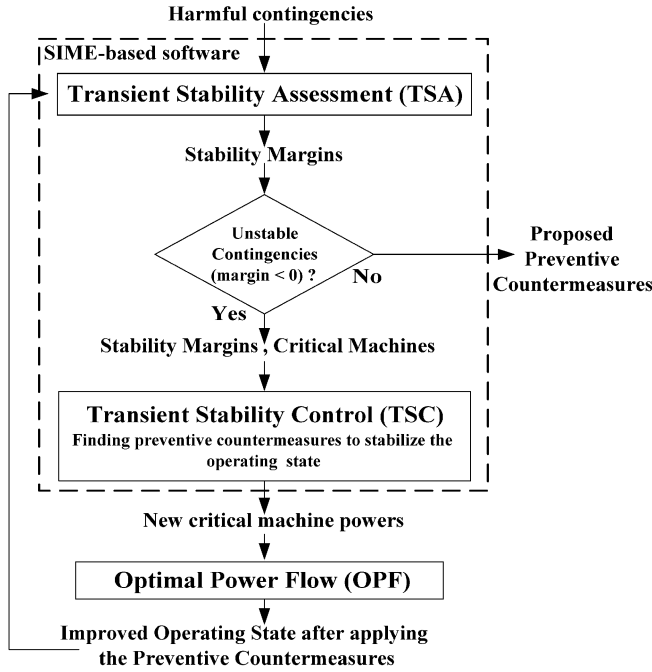


Fig. 7. Transient stability-constrained OPF (sequential approach).

- ii) This block computes the amount of power to be shifted from CMs, and sends this information to the OPF block, which compensates for this change by redispatching an almost equal generation on NMs.
- iii) The resulting new operating condition is sent to the TSA block, which tests its stability with respect to the contingencies of concern. If the case is found to be still unstable, the contingencies and their information (CMs and margins) are sent again to the TSC block; otherwise, the procedure stops.

D. Illustrations

The above transient stability-constrained OPF software is illustrated by simulating the Mexican interconnected system under a maximum demand case of June 2001 (27 450 MW). The system is modeled in detail with 154 machines, 2261 buses, and 3521 lines [5]. Of the 154 machines, 57 have a constant power output and 97 can be dispatched by OPF. The simulations use two base cases obtained by running a Newton Raphson ac OPF program, under two different objectives: minimum operating cost and maximum power transfer in dedicated tie lines.

For each one of these cases, applying the “n-1” criterion to the 400-kV bulk transmission system created 298 contingencies. The FILTRA software identified two harmful contingencies when the operating time of the protective devices is set to 115 ms. These contingencies have consistently been recognized to limit system operation.

In what follows, the most harmful contingency is stabilized by the transient stability-constrained OPF, under the two aforementioned objectives.

1) *Minimizing the Overall Operating Cost*: The first unstable case (most harmful contingency applied to the base case

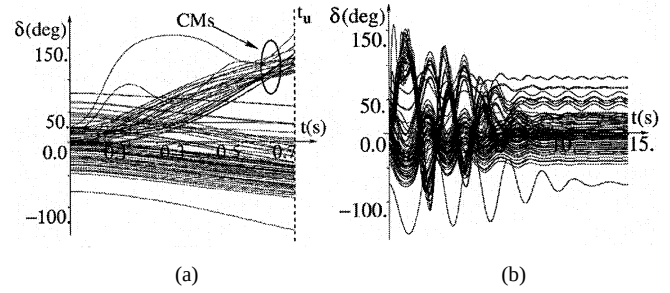
Fig. 8. Swing curves of the Mexican system corresponding to (a) the last unstable case ($P_c = 6162$ MW) and (b) the stabilized case ($P_c = 6059$ MW) of Table V.

TABLE V
PATTERN 1. CONTROLLING 40 CMs AND REDISPATCHING 57 NMs BY OPF TO MINIMIZE OPERATING COST

1	2	3	4	5	6
It. #	η (rad/s) ²	Nr of CMs	P_{c_k} (MW)	$P_{c_{k+1}}$ (MW)	sTDI (s)
1	-29.27	2	488	472	0.250
2	-6.809	48	6162	6059	0.797
3	0.000	/	6059	/	10.00

TABLE VI
SUMMARY OF THE BASE CASE AND OF THE THREE STABILIZING PATTERNS

1	2	3	4	5
Case	Cost (\$)	Losses (MW)	Line power (MW)	Total ΔP_c (MW)
0 Initial	5 629 257	767.8	230	0
1 Controlling 40 CMs and re-dispatching 57 NMs by OPF.	5 672 949	760.0	120	119
2 Controlling 8 CMs and re-dispatching 57 NMs by OPF.	5 665 088	758.9	126	109
3 Controlling 40 CMs and re-dispatching 57 NMs by LF.	5 678 855	762.8	126	109

set up by OPF with the objective to minimize the operating cost), has been stabilized according to the following three patterns.

- **Pattern 1**: Shifting generation from all 40 CMs proportionally to their inertia and redispatching the 57 NMs by OPF so as to minimize the total operating cost.
- **Pattern 2**: Shifting generation from the first eight CMs proportionally to their inertia and redispatching the 57 NMs by OPF so as to minimize the total operating cost.
- **Pattern 3**: Shifting generation from all 40 CMs proportionally to their inertia, and redispatching the 57 NMs using a load flow program.

Fig. 8 displays the multimachine swing curves of the last unstable and of the stable cases of pattern 1. There are 48 CMs and 106 NMs; of them, only 40 CMs and 57 NMs can be redispatched. The swing curves of Fig. 8(a) suggest that the CMs have almost the same degree of criticality (advancement). Table V describes the details of this stabilization procedure. The stabilization of the other patterns is very similar, except for the redispatch of NMs. Table VI gathers the values of the base case and of the three stabilization patterns in terms of operating cost (column 2), total power losses (column 3), power transfer in the interface line connecting Northern areas to the rest of

TABLE VII
CONTROLLING SIX CMS BY SIME & REDISPATCHING
91 NMS BY OPF TO MAXIMIZE POWER TRANSFER

1	2	3	4	5	6
It. #	η (rad/s) ²	Nr of CMs	P _{ck} (MW)	P _{ck+1} (MW)	sTDI (s)
1	-13.813	6	1974	1914	0.280
2	10.475	/	1914	/	10.00

TABLE VIII
COMPARISON OF THE INITIAL CASE AND THE ONE STABILIZED BY THE
TSC-OPF TECHNIQUE TO MAXIMIZE POWER TRANSFER

1	2	3	4	5
Case	Cost (\$)	Losses (MW)	Line power (MW)	Total ΔP_e (MW)
0 Initial (unstable)	6 280 781	838.8	233	0
1 Stabilized	6 257 818	803.2	233	60

the interconnected system (column 4), and amount of power to shift in order to stabilize the system (column 5).

These results show that pattern 2 is the best. They also show the advantages of OPF over power flow for redispatching system machines (compare patterns 1 and 2 with pattern 3).

2) *Maximizing the Power Transfer on an Interface:* A second base case of the Mexican system was obtained by using OPF to maximize the power over the interface lines connecting the Northern areas with the rest of the interconnected system. This interface is composed of two long 400-kV lines.

The most harmful contingency for this transient stability-unconstrained operating condition has six CMs belonging to the same power plant. To stabilize the system, the generation of these CMs was shifted according to the transient stability control technique; the 91 NMs were redispatched using OPF so as to maximize the interface power.

Table VII describes the stabilization process, while Table VIII shows an interesting result, namely, that the transient stability-constrained OPF was able to stabilize the system without decreasing the maximum interface power transfer and without additional cost.

3) *Discussion:* In the above sequential approach to transient stability-constrained OPF, the CMs' generation has been redispatched by SIME and not by OPF, in order to let OPF meet its main objective function only (minimizing operating cost or maximizing power transfer). But it would be also possible to rely on OPF for redispatching generation among CMs as well.

More generally, let us note again that the approach is very flexible, and may match different market rules. In particular, it may help the system operator solve congestion management problems in the balancing market, 1 h ahead of real-time operation. Market issues are addressed in [1, Sec. V].

V. CONCLUSIONS

In this paper, a transient stability control methodology has been proposed for the automatic design of preventive countermeasures.

The simulations conducted on the EPRI 88-machine system and the Mexican 154-machine system illustrated its specifics and highlighted its performance.

The resulting software has been shown to combine simplicity, accuracy and, in addition, to be

- flexible with respect to power system modeling, contingency type and scenario, instability modes;
- robust with respect to power system topology, operating conditions, contingency severity;
- reliable with respect to advocated solutions;
- able to provide a large variety of near-optimal solutions;
- able to easily comply with online computing requirements;
- easy to combine with a standard OPF software, yielding a transient stability-constrained OPF.

This transient stability-constrained OPF approach has been applied to the Mexican interconnected power system, and two issues of current concern in power system operation have been addressed.

- First, a security-constrained redispatch technique, which minimizes the overall cost of security control. This technique can be applied to mandatory or optional balancing (or congestion management) markets. Indeed, it has been shown that the problem can be solved by acting only on a subset of the system machines participating in the dispatch. This technique is also suitable for real-time operation, in which the system operator could transparently justify the decisions for redispatching the different available resources while minimizing the cost of security control.
- Second, a redispatch technique, which maximizes security-constrained transfer limits. Neglecting the costs during the transfer limit calculations would reinforce the requirement for a nondiscriminatory treatment of all generators, taking into account only physical (thermal, voltage, and stability) limits.

All in all, the focus of this paper has been on methodological matters rather than applications to various market models. However, the obtained results suggest that the derived methods are fully flexible to accommodate to various market models.

Finally, we note that the proposed transient stability control technique is the prerequisite to an alternative approach, complementary to the purely preventive one, namely, the open-loop emergency control developed in the companion paper [1].

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